INCH-POUND
MIL-W-16878/8B(NAVY)
11 September 1992
SUPERSEDING
MIL-W-16878/8A(NAVY)
10 August 1981

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, SILICONE RUBBER INSULATED, 200°C, 1000 VOLTS

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-16878.

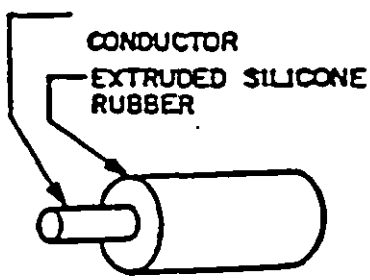


FIGURE 1. Construction.

AMSC N/A FSC 6145
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TABLE I. Construction details.

PIN 1/	Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material	Coating		Min	Max
M16878/8-BEB*	24	7 X 32	Copper	Silver	0.024	0.083	0.091
M16878/8-BFB*	22	7 X 30	Copper	Silver	.030	.089	.097
M16878/8-BGB*	20	7 X 28	Copper	Silver	.038	.097	.105
M16878/8-BGC*	20	10 X 30	Copper	Silver	.038	.097	.105
M16878/8-BHB*	18	7 X 26	Copper	Silver	.048	.107	.115
M16878/8-BHD*	18	16 X 30	Copper	Silver	.048	.107	.115
M16878/8-BJE*	16	19 X 29	Copper	Silver	.057	.116	.124
M16878/8-BJF*	16	26 X 30	Copper	Silver	.057	.116	.124
M16878/8-BKE*	14	19 X 27	Copper	Silver	.072	.161	.171
M16878/8-BKH*	14	41 X 30	Copper	Silver	.072	.161	.171
M16878/8-BLE*	12	19 X 25	Copper	Silver	.091	.180	.190
M16878/8-BLJ*	12	65 X 30	Copper	Silver	.091	.180	.190
M16878/8-BMG*	10	37 X 26	Copper	Silver	.111	.200	.210
M16878/8-BMK*	10	105 X 30	Copper	Silver	.120	.200	.210
M16878/8-BNL*	8	133 X 29	Copper	Silver	.169	.281	.307
M16878/8-BPL*	6	133 X 27	Copper	Silver	.213	.326	.351
M16878/8-BRL*	4	133 X 25	Copper	Silver	.269	.382	.407
M16878/8-BSL*	2	133 X 23	Copper	Silver	.335	.448	.473
M16878/8-BTM*	1	259 X 25	Copper	Silver	.378	.519	.550
M16878/8-BUM*	0	259 X 24	Copper	Silver	.424	.565	.596
M16878/8-BWM*	00	259 X 23	Copper	Silver	.477	.618	.649
M16878/8-BYM*	000	259 X 22	Copper	Silver	.533	.694	.725
M16878/8-BZM*	0000	259 X 21	Copper	Silver	.601	.742	.773

1/ PIN stands for part or identifying number.

ADDITIONAL REQUIREMENTS:

Visual and mechanical examination: Required.
Spark test: 5.0 kV.
Impulse dielectric test: 8.0 kV.
Dielectric withstanding voltage: 3.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C.
K = 1,000.
D = Maximum average diameter of finished wire.
d = Conductor diameter.

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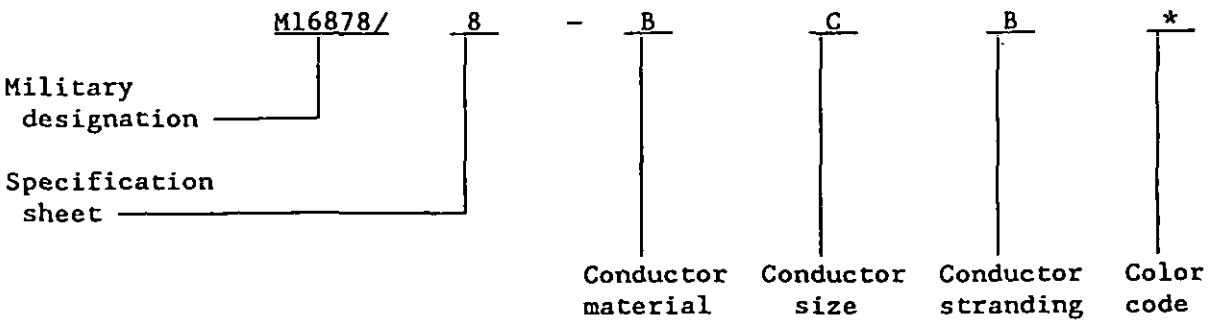
Cold bend: Condition 4 hours at minus 55 ± 1°C (see table II).

TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (maximum diameter in inches)
24, 22	1
20 through 12	2
10 through 6	3
4, 2	4.5
1, 0	6
00, 000, 0000	10

Concentricity: 70 percent (minimum).
Surface resistance: Not required.
Wrap back: Not required.
Heat resistance: Condition at 250°C.
Flammability: Meets requirements of UL VW1 Flame Test.
Toxicity: Meets requirements of concentration levels specified in MIL-E-917.
Shrinkage: Required.
Heat aging: Not required.
Insulation tensile strength: 700 pounds force per square inch (minimum).
Insulation elongation: 125 percent (minimum).
Marking and stripe durability: Required.
Fungus resistance: Required.
Maximum dc resistance of finished wire: See table I of MIL-W-16878.
Identification of product: Required for wire sizes 22 AWG and larger.

PIN example (see MIL-W-16878):



Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6145-N330-08)

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